

IB MYP
Summative Assessment - 2 (2026-27)

Students Name:	Grade & Sec: MYP-1	Date: 31-08-2026
Subject: Integrated Sciences	Task Duration: 1 hour	Level Obtained- Criterion-D-
Unit Title : Unit-2-What Changes {Matter and materials}		
Criterion-D-Reflecting on the impacts of science		
ATL Category, Cluster, Strand: Critical Thinking – analyze information and evaluate solutions Learner Profile: Thinker		

CRITERION- :

Achievement Level	Level Descriptor	Task-Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard
1-2	The student should be able to: i. state the ways in which science is used to address a specific problem or issue ii. state the implications of using science to solve a specific problem or issue, interacting with a factor iii. apply scientific language to communicate understanding with limited success iv. document sources, with limited success .	The student is able to: i. State how solar still uses science to help address water scarcity. ii. State an implication of using a solar still and make a simple connection to an environmental, economic or social factor. iii. Apply scientific language with limited success. iv. Document sources with limited success.

3-4	The student should be able to: i. outline which science is used to address a specific problem or issue ii. Outline the implications of using science to solve a specific problem or issue, interacting with a factor iii. Sometimes apply scientific language to communicate understanding iv. Sometimes document sources correctly	The student is able to: i. Outline how a solar still uses evaporation and condensation to help address water scarcity. ii. Outline an implication of using a solar still and connect it to an environmental, economic or social factor. iii. Sometimes apply appropriate scientific language. iv. Sometimes document sources correctly.
5-6	The student should be able to: i. summarise how science is used to address a specific problem or issue ii. Describe the implications of using science to solve a specific problem or issue, interacting with a factor iii. Usually apply scientific language to communicate understanding clearly and precisely iv. Usually document sources correctly	The student is able to: i. Summarize how a solar still uses scientific understanding of changes of state to address water scarcity. ii. Describe the implications of using a solar still and how they interact with an environmental, economic or social factor. iii. Usually apply scientific language clearly and precisely iv. Usually document sources correctly.
7-8	The student should be able to: i. Describe how science is used to address a specific problem or issue	The student is able to: i. Describe clearly how scientific understanding of evaporation and condensation is applied in a solar still to address water scarcity.

	ii. Discuss and analyse the implications of using science to solve a specific problem or issue, interacting with a factor iii. Consistently apply scientific language to communicate understanding clearly and precisely iv. Document sources completely	ii. Discuss and analyse the implications of using a solar still, considering an environmental, economic or social factor, and use evidence to reach a justified recommendation. iii. Consistently apply accurate scientific language clearly and precisely. iv. Document sources completely.
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Criterion-D-Reflecting on the impacts of Science

Water Cycle, Changes of State and the Solar Still

THE CONTEXT

Water continuously moves between the Earth's surface and the atmosphere through the **water cycle**. Changes of state such as **evaporation** and **condensation** are important parts of this cycle. Some places receive very little rainfall. People living in these areas may experience **water scarcity**, which means there may not be enough clean water for drinking and everyday needs. Scientists and engineers use their understanding of **changes of state** to develop technologies that can help people obtain cleaner water. One example is a **solar still**.

You can choose any other solution

THE PROBLEM

A small community experiences long periods with very little rainfall. The people need more clean water for drinking and everyday activities. Scientists have suggested using a **solar still** to help the community obtain cleaner water.

The community wants to know:

Should we use solar stills to help address our water shortage?

You have been asked to investigate the science and make a recommendation.

RAFT

R – Role: Young Water Scientist

A – Audience: Local Community and Environmental Committee

F – Format: Scientific Report / A3 Scientific Poster

T – Topic: Can solar still help a community experiencing water scarcity?

HOW DOES A SOLAR STILL WORK?

A solar still uses **energy from the Sun** and **changes of state** to obtain cleaner water.

Study the solar-still diagram provided carefully.

Use the information and diagrams provided to complete the following task.

YOUR TASK

Prepare a **scientific report or A3 scientific poster** for the community.

Your work should include the following sections.

PART 1 – WHAT IS THE PROBLEM?

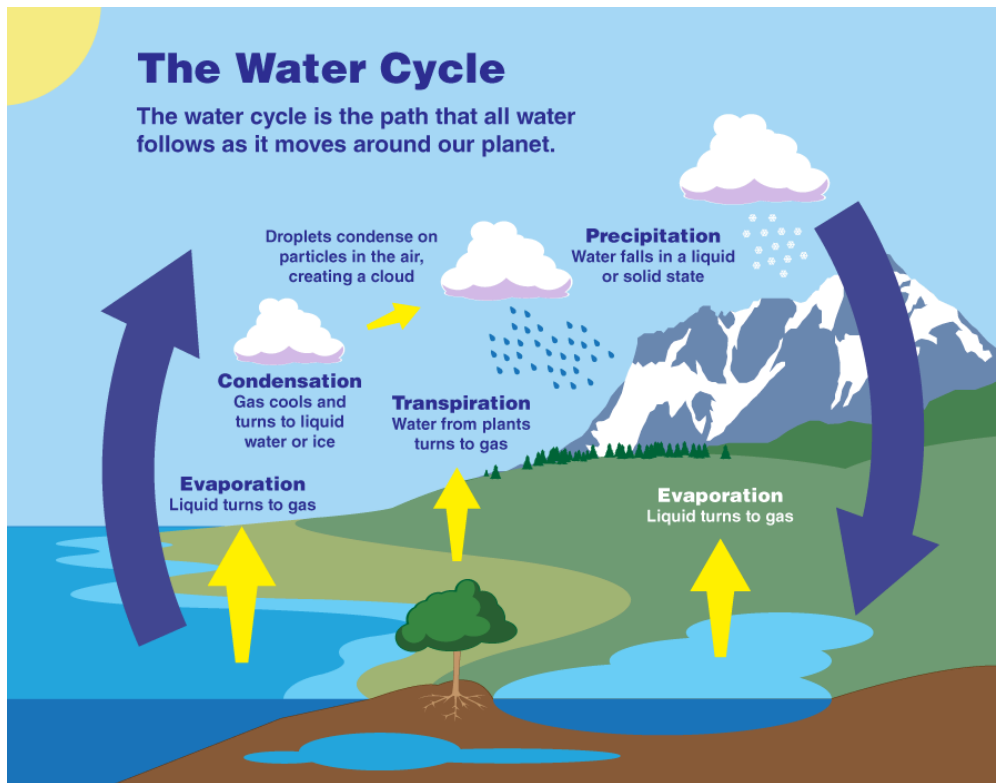
In your own words, **state the problem** faced by the community.

Think about:

- rainfall
- availability of clean water
- people's everyday needs
- water scarcity

PART 2 – HOW DOES SCIENCE HELP?

Study the **water-cycle diagram** and the **solar-still diagram** provided.



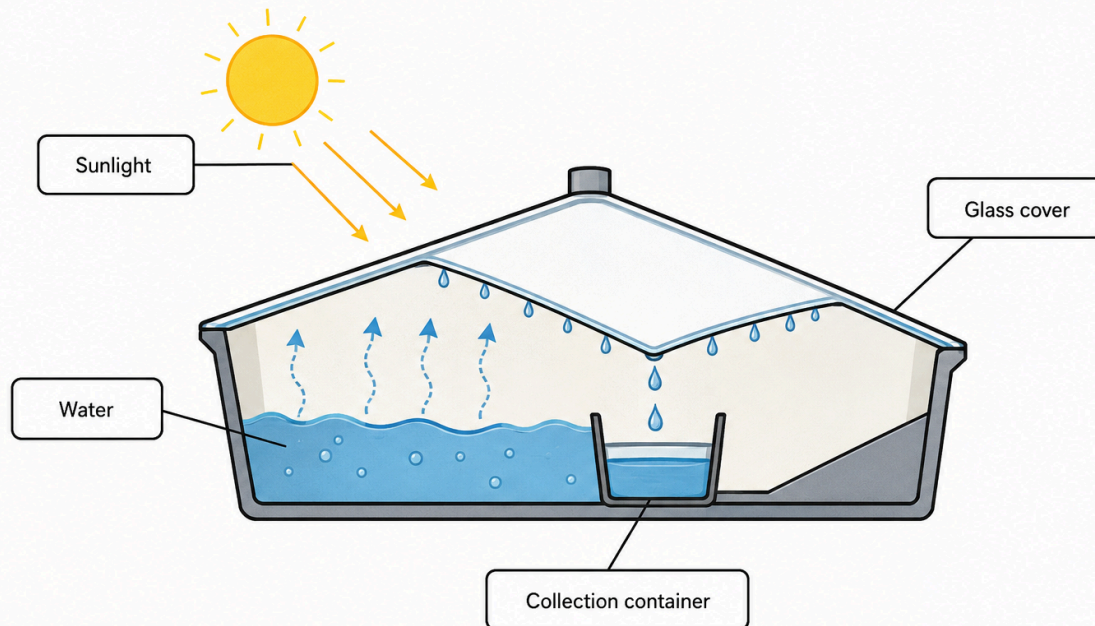
a. Identify

Identify the change of state in which:

1. liquid water changes into water vapour.
2. water vapour changes back into liquid water.

SOLAR STILL

Study the diagram below carefully and use it to answer the questions.



Research how solar still works.

A solar still:

- uses **sunlight as its main energy source**
- can be used to obtain **cleaner water from salty or impure water**
- usually produces **a limited amount of water at a time**
- works best when there is **strong sunlight**
- requires materials and money to **build**, but needs little energy to operate

Research how solar still works.

b. Connect the science

How are these same two changes of state used in a **solar still**?

c. Summarize

Summarize how a solar still uses these changes of state to help address water scarcity.

Think about:

- What happens when the Sun heats the water?
- Which change of state happens first?
- What happens to the water vapour when it reaches the cooler surface?
- How is the liquid water collected?
- How can the collected water help the community?

PART 3 – ONE BENEFIT

Describe ONE benefit of using a solar still.

You could think about:

- obtaining cleaner water
- using energy from the Sun
- reducing the need for electricity
- usefulness in places where clean water is limited

Sentence starter

One benefit of using a solar still is _____ because _____.

PART 4 – ONE LIMITATION

Describe ONE limitation of using a solar still.

You could think about:

- the amount of water produced
- dependence on sunlight
- time needed to collect water
- cost of building the system

Sentence starter

One limitation of using a solar still is _____ because _____.

PART 5 – THINK ABOUT THE IMPLICATIONS

An **implication** is an effect or consequence that may happen when science is used.

A benefit or limitation tells us something about the **solar still itself**.

An implication asks us to think about:

How could using solar still affect people, money or the environment?

Choose any **two factors**.

OPTION A – ENVIRONMENTAL FACTOR

Think about:

- Does a solar still require electricity while operating?
- Does using energy from the Sun help reduce environmental impact?
- Are materials needed to build the solar still?

OR

OPTION B – ECONOMIC FACTOR

Think about:

- Does it cost money to build a solar still?
- Does it cost much to operate?
- Could using solar energy reduce some operating costs?

OR

OPTION C – SOCIAL FACTOR

Think about:

- Could **solar still** improve people's access to cleaner water?
- How could this help families?
- Could it be useful in communities where clean water is difficult to obtain?

Write about your chosen factors:

The factors I chose are _____.

One positive implication is _____ **because** _____.

One possible negative implication is _____ **because**
_____.

Overall, this means _____.

PART 6 – MAKE YOUR RECOMMENDATION

Now use the scientific information you have considered to make a decision.

Would you recommend using solar stills in this community?

Give **scientific reasons and evidence** for your recommendation.

Your recommendation should consider:

- how the solar still works
- one benefit
- one limitation
- the implication you discussed

You may use this structure:

I recommend / do not recommend using solar stills because
_____.

The scientific processes of _____ **and** _____
allow the solar still to _____.

One important benefit is _____.

However, one limitation is _____.

Considering the _____ **factor, an important implication is**
_____.

Therefore, I think the community should / should not _____
because _____.

Remember

Your recommendation should be based on **scientific evidence**, not only your personal opinion.

PART 7 – SCIENTIFIC COMMUNICATION AND SOURCES

Scientific Vocabulary

Use appropriate scientific vocabulary throughout your work.

Try to correctly use words such as:

water cycle • evaporation • condensation • liquid • gas • water vapour • thermal energy • change of state • water scarcity • solar energy • collection

Sources

Use at least **TWO reliable sources** to complete your work.

Write your sources at the end of your report/poster.

Bibliography

Source 1: _____

Source 2: _____

FINAL CHECK BEFORE SUBMITTING

Make sure your work:

- explains the community's problem
- shows how science is used in a solar still
- uses evaporation and condensation correctly
- includes one benefit and one limitation
- discusses positive and negative implications linked to **one factor**
- gives a scientifically supported recommendation
- uses appropriate scientific vocabulary
- includes at least two reliable sources